Project report

Lumina Fluids Bay

Project type	Status	Role	Dates
Hardware	In progress	Vehicle Fluids Lead	Jun - Present 2026

Overview

Development of liquid bipropellant (Ethanol x Nitrous), regeneratively cooled launch vehicle using the UL-1 Engine. Responsible for the end-to-end development of propellant feed systems, including line sizing, P&ID, fluids bay packaging, build sequencing, and assembly.

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01 Line Sizing

The Lumina vehicle burns ethanol fuel with nitrous oxide oxidizer. Feed lines were sized by balancing the design considerations below, ordered from most to least important.

1. **Pressure loss.** The fuel feed line may lose up to 25 psi and the oxidizer feed line up to 50 psi. Because the nitrous is stored at saturation, the oxidizer line is designed to keep ΔP as low as possible.
2. **Velocity.** Most relevant to the oxidizer line. As a proxy for two-phase effects on mass flow, density and ΔP, lines were sized to bring flow velocity as close to 10 m/s as possible.
3. **Hoop stress and factor of safety.** Feed lines see 750 psi and 1200 psi. Material and wall thickness must give a factor of safety of 2 against von Mises hoop stress at 1200 psi.
4. **Tube bendability.** Selections respect a wall factor (OD / wall) under 15 and the 1/16 to 1/28 rule, to keep tubes from crumpling when bent.
5. **Cost.** 316 SS generally costs more than 304 SS. Cost is a softer constraint, and budget is better spent on fittings, which are usually the largest expense.

Relations used in the trade studies

Flow velocity comes from the mass flow and the tube's inner area. Hoop stress uses the inner diameter, and the factor of safety compares the material's yield strength to that stress at 750 psi and 1200 psi. Pressure loss is given over a 3 ft run of tube.

Flow velocity

$$V = \frac{\dot{m}}{\rho A}$$

Flow area

$$A = \frac{\pi ID^2}{4}$$

Hoop stress

$$\sigma_h = \frac{P\,ID}{2t}$$

Factor of safety

$$SF = \frac{\sigma_y}{\sigma_h}$$

Variables

v	Flow velocity	m/s	σ_h	Hoop stress in the tube wall	psi
$\dot{m}$	Mass flow rate	kg/s	P	Internal pressure, 750 or 1200	psi
ρ	Propellant density	kg/m^3	t	Wall thickness	in
A	Inner flow area of the tube	m^2	σ_y	Yield strength of the tube material	psi
ID	Tube inner diameter	in	SF	Factor of safety against yield	none

Propellant properties at 70 °F

	Ethanol (fuel)	Nitrous oxide (oxidizer)
Density	789 kg/m^3	776 kg/m^3
Viscosity	0.001158 Pa·s	6.6785 × 10^{-5} Pa·s

Tube materials

	Yield strength
316 SS	42,100 psi
304 SS	31,200 psi

Mass flow rates

	Value
Engine $\dot{m}$	1.98 kg/s
Oxidizer $\dot{m}$	1.42 kg/s
Fuel $\dot{m}$	0.65 kg/s
Regen mass fraction	0.33

Fuel line trade study (ethanol)

		> 0.035 in		Minimize		< 25 psi		> 2	> 2
Material	**OD (in)**	**Wall (in)**	**ID (in)**	**Cost per 3 ft**	**Velocity (m/s)**	**ΔP over 3 ft (psi)**	**Yield strength (psi)**	**SF at 750 psi**	**SF at 1200 psi**
316 SS	1/2	0.035	0.430	$65.74	10.41	10.16	42,100	9.14	5.71
316 SS	1/2	0.049	0.402	$49.16	11.92	14.08	42,100	13.68	8.55
304 SS	1/2	0.035	0.430	$27.07	10.41	10.16	31,200	6.77	4.23
316 SS	3/8	0.035	0.305	$27.54	20.70	54.09	42,100	12.88	8.05
316 SS	3/8	0.028	0.319	$31.73	18.93	43.43	42,100	9.85	6.16

Oxidizer line trade study (nitrous oxide)

		> 0.035 in		Minimize	< 13 m/s	< 50 psi		> 2	> 2
Material	**OD (in)**	**Wall (in)**	**ID (in)**	**Cost per 3 ft**	**Velocity (m/s)**	**ΔP over 3 ft (psi)**	**Yield strength (psi)**	**SF at 750 psi**	**SF at 1200 psi**
316 SS	1/2	0.035	0.430	$65.71	19.53	24.01	42,100	9.14	5.71
316 SS	3/8	0.035	0.305	$27.54	38.00	140.00	42,100	12.88	8.05
316 SS	7/16	0.049	0.340	$62.12	31.20	80.00	42,100	16.18	10.11
316 SS	1/2	0.049	0.402	$49.16	22.35	33.90	42,100	13.68	8.55
304 SS	1/2	0.035	0.430	$27.07	19.53	24.01	31,200	6.77	4.23
304 SS	1/2	0.028	0.444	$27.10	18.31	20.38	31,200	5.25	3.28
304 SS	5/8	0.049	0.527	$39.87	13.00	8.50	31,200	7.74	4.83
304 SS	5/8	0.035	0.555	$33.17	11.72	6.53	31,200	5.25	3.28
304 SS	3/4	0.109	0.532	$90.41	12.76	8.10	31,200	17.05	10.65
304 SS	5/8	0.016	0.593	$97.72	10.30	4.67	31,200	2.24	1.40

Other considerations

Engine clearances required the fuel feed connection to step down to 3/8 in. A 3/4 in oxidizer line gave the best results for ΔP and flow velocity, but it was not feasible to bend or to package alongside the fuel and oxidizer valves.

Selected lines

Fuel (ethanol): 6061 aluminum, 1/2 in OD × 0.035 in wall
Oxidizer (nitrous oxide): 6061 aluminum, 5/8 in OD × 0.035 in wall